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1	30/04/2018	Creation (Grouping and Simplification of the Technical Annexes to the Requirement Specifications)
Edition No.	Date	Motive for the alterations in the last three editions

This document describes the technical requirements allowing for the standardisation of monitoring, service and maintenance of Work Equipment (WEq) at the BOSCH plant in Rodez (RzP).

In order to prevent any risk of obsolescence, the used equipment must be available during the active marketing period for a minimal duration of 5 years (after equipment acceptance). The supplier will use **commercially available** equipment. If not possible, he must get the agreement of BOSCH for operating special equipment.

The spare parts list with references and manufacturers must be delivered to Bosch Company; and in case of suppliers' special parts, their availability conditions, especially their prices and delivery lead times, must be specified.

If the ordered installation is "identical" to another existing one in the plant, the supplier must specify all technological improvements from one installation to the other.

The WEq must be in conformity with the applicable principles stated in:

- Directives
  2006/42/EC (Machinery),
  2014/35/EU (Low Voltage),
  2014/30/EU (Electromagnetic Compatibility) and
  2014/68/EU (Pressure Equipment),
- National standards and specifications used (French Labour Code, Special Decrees, Specific Directives).

The manufacturer will apply the valid environmental regulations applicable to the provided WEq and at least comply with

- European Directive 2009/125/EC (21rst October 2009) from the European Parliament establishing a framework for the setting of eco-design requirements for "energy-related products".

- Decree from 02/02/98 about the taking and consumption of water as well as about all kinds of emissions from facilities classified for environmental protection subject to authorisation (ICPE)

- Decree from 30/06/2006 relating to surfaces treating facilities subject to authorisation as defined in section 2565 of the nomenclature of classified facilities.

1. Mechanical and Hydraulic Standard

All equipment parts and components must be protected against corrosion with coatings adapted to the environment and the process of the WEq.

The machine must be equipped with anti-vibrating adjustable legs or shoes.

The peripheral devices (loading, etc…), especially the hydraulic system, must be positioned so as not to transmit any vibration to the equipment.

Wear and spare parts: they must

- have a simple design,
- resist wearing (for example, if not necessary, do avoid the use of aluminium toothed pulleys),
- be accessible for checking and/or cleaning,
- be easily interchangeable (the supplier will guarantee as a rule that the mounting/dismounting of a part doesn't take more time than 30 minutes).

Accessibility to the different equipment components must be guaranteed to provide safety during repairs; if need be, arrange an appropriate hoisting and handling device matching the equipment.

Handling and hoisting of sub-assemblies:

Plan enough anchoring points to enable the hoisting in only 1 point (this has to be specified on the handling drawing of the equipment).

Easy accessible collecting points are to be planned in order to perform the analyses on the different discharges in the air and on the water rejects.

The WEq must be tight and designed to avoid any run-offs on the ground as well as to lessen the risk of external pollution to the components; this requirement will be checked, among others, during the acceptance procedure of the WEq.

The only machines to be equipped with retention containers will be those performing cleaning, stripping and degreasing.
All access points where the employees have to operate at least once a week must be equipped with doors having safety devices. These accesses must have identical protection levels (for example, no combination of bolt-on locking systems and locking systems with electrical safety devices).

All doors or casing glasses must be made of polycarbonate or safety glass.

For all vertical casings or covers, plan a mechanical locking system in addition to the cylinder, in order to provide an optimal safety.

Notes:

- Pipework:
  Favour rigid tubes
  In any case, adjust the pipework type to the type of fluid.
  Fittings with 24° cone, Series L or S (depending on pressure). If other fittings types are required inform the BOSCH Company.
- Valves:
  Type: Ball valve up to DN 50, butterfly valve or similar above DN 50.
- Pressure gauge:
  Favour the brand WIKA
  Stainless steel
  Dial diameter: 63 mm mini
  Oil bath with pressure intake
- Check valve:
  Favour the brand YORK
- Pumps:
  Favour the brand KSB or GRUNDFOSS
  Flanged mounting with isolation valve upstream and downstream
  Pressure test intakes by differential pressure gauge.

The temperature regulation must be adjusted to the fluid and the need.

The cutting oil collecting tank is provided by the BOSCH Company if nothing else is specified in the Requirements Specification (in this case, make sure that level sensors are installed in order to avoid any risk of overflow).

- **Fluid tanks:**

For each compartment, provide for easily accessible taking valves, as well as for the system at the pressure outlet of the tank.
Drain port at the lowest point of each zone (minimal diameter 60 mm). The bottom must be tilted towards the drain.
A lateral trapdoor enables the cleaning of the container bottom and must provide access to all tank points.
Electrically triggered level and temperature detectors with indicator light are to be installed. The temperature regulation must be adjusted to the fluid and the need.

- **Hydraulic unit:**

Definition of equipment and installation wiring must follow the state of the art so as to guarantee the safety and the reliability of the system.

To take into account the energy efficiency and eco-design criteria, it is advisable to use preferably an auto-regulated hydraulic power unit, high efficiency engines and hydraulic elements attuned to the appropriate needs regarding flow and pressure so as to guarantee a proper functioning in all operating modes of the installation.

To be able to analyse the system's functioning, "mini-mess"-type pressure control points must be installed on the discharge side of the pump, at the inlets and returns of the collection lines.

Equipment brand preference is Rexroth, favouring the newest generation of products. Some applications may require other equipment: in this case the prior approval of BOSCH Company is mandatory.

Except in specific cases, the stacking of components must not exceed 3 elements.

The choice of implantation must be submitted to the BOSCH Company for approval before building the equipment.

Oil filtration level must be adjusted to the requirements for the components mounted into the system, according to the valid standards.

Hydraulics

Brand preference is Rexroth.

- Pumps

The pump type must be determined according to the use.

- Distributors:

They must be equipped with connectors and visualisation LEDs for spool position monitoring, and they must be accessible. Favour the distributors Series NG6.

- Pressure switches: favour the sub-plate mounting.
- Flow throttling and regulation valves: favour the sub-plate mounting.
- Pressure regulators: favour the sub-plate mounting.
- Modular system assembling: favour Series NG6 and NG10.

- Accumulators:

  NBR bladder accumulators with volume adjusted to the system; however the compatibility of bladder material with the used fluid must be guaranteed. For a volume of 4 litres or more, the accumulators must be equipped with a pressure gauge in order to continuously check the bladder inflation pressure.

  They must be necessarily equipped with safety blocks allowing for the automatic flow to the tank in case of machine shutdown.

  Accumulator manufacturing year should not predate the date of equipment order.

  Observe the valid regulation: Directive 2014/68/EU of the European Parliament and of the Council of 15 May 2014 on the harmonisation of the laws of the Member States relating to the making available on the market of pressure equipment.

- Hoses:

  Hoses manufacturing date must be specified on each hose.
  Mounting and protection of the hydraulic hoses must be executed according to the valid safety standards.
  The flexible hoses conveying fluids, especially under pressure, must be able to withstand the foreseen internal and external stresses and must be firmly attached and/or protected in order to ensure that no risk is posed by a rupture. The choice of solutions is limited to flow reducing protection sleeves at the ends as well as anti-whip cables secured to a stationary element of the equipment.

  Caution: Identify with permanent signs all the elements (actuators and hoses) remaining under pressure after a power switch-off (emergency stop).

Mechanical and hydraulic equipment to be favoured:

EQUIPMENT TYPE	RECOMMENDED BRANDS
HYDRAULIC CYLINDER	REXROTH
	HANCHEN
PNEUMATICALLY DRIVEN PUMP	DEPA
SCREW SLUDGE PUMP	PCM MOINEAU
CENTRALISED GREASING	VOGEL / SKF
	HYDAC
PVC VALVES - PNEUMATICALLY ACTUATED VALVES	GF
	GEMU
REGULATION COMPONENT SERVO VALVE	BOSCH REXROTH
HORIZONTAL STAINLESS STEEL MONOCELLULAR PUMP	KSB
VERTICAL STAINLESS STEEL MULTICELLULAR PUMP	GRUNDFOSS / KSB
2/2 3/2 PNEUMATICALLY ACTUATED VALVES STAINLESS STEEL	BURKERT / ASCO / GEMU
SCREW HYDRAULIC PUMP	KNOLL
PISTON HYDRAULIC PUMP	BOSCH REXROTH
PINION HYDRAULIC PUMP	BOSCH REXROTH
COLLECTING PUMP	KSB / GRUNDFOSS
VERTICAL MULTICELLULAR PUMP	GRUNDFOSS
COAXIAL ELECTROVALVE COOLING	COAX / ASCO
COMPONENT PROPORTIONAL	BOSCH REXROTH
	HERION
COAXIAL ELECTROVALVE	COAX / ASCO
PNEUMATIC HANDLING COMPONENTS	SCHUNK SMC / AFAG
	BOSCH REXROTH
	AVENTICS
	HEINZ MAYER
GUIDING (RAIL AND MONORAIL)	SCHNEEBERGER / IGUS
	INA / BOSCH REXROTH
REDUCER MOTOR / BRAKE MOTOR	LEROY SOMMERS
	SEW
	LENZE
PLANETARY REDUCER	ALPHA / BOSCH REXROTH

➢ Greasing

Employed oils and greases must be selected in the following table. Any incompatibility has to be justified to the BOSCH Company and, in any case, oils and greases supplied by Shell have to be favoured.

BOSCH RODEZ approved oils and greases
03 / 2018

	COMME	LUBRICANT TYPE
SHELL	TELLUS S2 M 46	Mineral hydraulic oil
SHELL	TELLUS S2 M 68	Mineral hydraulic oil
SHELL	TELLUS S2 M 32	Mineral hydraulic oil for greasing circulating system (air)
SHELL	TELLUS SX 32	Synthetic hydraulic oil
SHELL	TELLUS S2 MX22	Oil for hydrostatic guiding plate
SHELL	TIVELLA S4 WE 220	Extreme pressure synthetic oil (reducers)
SHELL	TONNA S3 M68	Mineral oil for sliding guides, reducers and hydraulics
SHELL	GLISS GO 150	Mineral oil for sliding guides, reducers and hydraulics
SHELL	MORLINA S2 BA 5	Fine oil for bearings, spindles, high speed bearings
SHELL	MORLINA S2 BA 10	Fine oil for bearings, spindles, high speed bearings
SHELL	OMALA S2 G 220	Extreme pressure mineral oil (reducers)
SHELL	OMALA S2 G 460	Extreme pressure mineral oil (reducers)
SHELL	OMALA S4 GX 150	Synthetic gear oil (HP pumps)
SHELL	OMALA S4 GX 320	Synthetic gear oil
SHELL	OMALA S2 G 460	Gear oil
LEYBOLD	LEYBONOL LV 130	Vacuum pump oil
SHELL	GADUS S2 V 220 2	Semi-fluid gear grease
SHELL	GADUS S2 V 100 2	Grease for axes and electrical cylinders, robots
SHELL	GADUS S2 V 220 00	Semi-fluid gear grease
KLUBER	ISOFLEX NBU 15	Extreme pressure synthetic grease
KLUBER	ALTEMP Q NB 50	Lubricating and assembly paste for mandrels

2. Pneumatic Standard

2.1 General information about pneumatics.

Taking into account the energy efficiency and eco-design criteria, use and consumption of pressurised air must be adjusted.
For this purpose, pneumatic elements must be attuned to the appropriate needs regarding flow and pressure so as to guarantee a proper functioning in all operating modes of the installation.

Implementation of components and wiring must be executed in order to avoid any risk of air leakage.

Use of pressurised air for the blowing must be limited to what is strictly necessary. Blowing time must be limited only to the necessary cycle sequences with automatic switch-off functions programmed in the NC on outputs such as “end-of-cycle”.

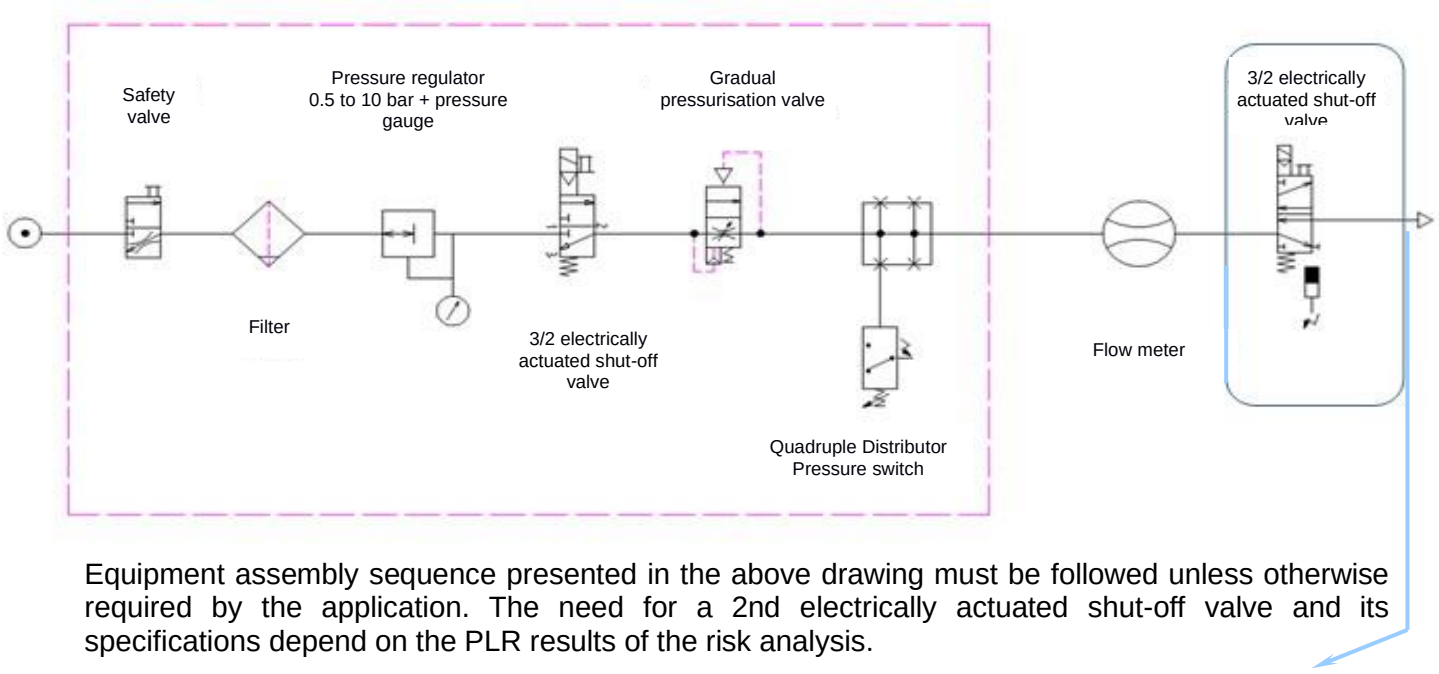
In the context of the “Energy Assessment” of means of production, the WEq must be equipped with a measuring device so as to collect information on the pneumatic energy consumptions, at a given time and cumulated, in Nm^3.

These measurements will be entered in one page of the Human/Machine Interface (HMI).

2.2 Pneumatics

Defined equipment brand to favour is Aventics.
Any use of equipment not described as a standard hereafter requires the prior approval of the BOSCH Company.

Pneumatic supply:



Equipment assembly sequence presented in the above drawing must be followed unless otherwise required by the application. The need for a 2nd electrically actuated shut-off valve and its specifications depend on the PLR results of the risk analysis.

Air-treatment system:

Use the Series AS2 ¼' and AS3 ¾' according to the flow needed in the installation.
The system comprises:



- A manually operated switch-off valve 3/2

Padlock locking is mandatory.



- A filter

The plant network has a filtration degree of 25 µm, however a filter adjusted to the requirement level of the equipment must be installed downstream in order to protect the installation.

Anyway a filter is necessary in case of a network failure.

The tank draining must be manual.

- A pressure regulator

A needle pressure gauge must be installed on the regulator body and it must have adjustable graduations allowing for the visualisation of mini/ maxi operating pressures.

(Operating pressure must be specified in the user manual of the station)





- An electrically actuated pressurisation unit

It comprises an electrically actuated 3/2 valve with a gradual pressurisation element for the installation.

If the installation is without electricity, this device will not be installed and will be replaced by a pneumatic actuated valve.



- An air distributor



- A pressure sensor mounted on the front part of the air distributor. Operating pressure range min 0 / maxi 10



Overpressure protection: 15 bar

Digital output signals: 2 outputs, PNP, NPN, Push–pull connector: threading G1/4

The device characteristics can be modified if the installation requires it.

If the installation is without electricity, this device will not be installed.

- A flow meter



Aventics AF flow meter mounted on the air-treatment system, or, IFM SD flow meter installed on the pipework for main supply of pressurised air.

Distributors and accessories:

- Mechanically actuated distributors

Series to be selected: AP or ST

- Pneumatically actuated distributors

Series to be selected: ISO1 or ST

- Electrically actuated distributors

Series to be selected:

  - Either HF02 or HF03

The choice between HF03 and HF02 depends on the flow needed to meet the requirements for operating the installation.
For the definition of distribution islands (bus), see the details in the electrical section of this standard.

  - Or ISO1 and Wv02 for sub-plate mounted or individual distributors.

Note: The distributors' manual commands should not be lockable.

- Position of the distribution blocks:

The distribution blocks must be positioned so as to guarantee easy access for the maintenance personnel, outside of the machine housing.
Special attention will be given to the placement and allocation of the distributors so as to limit as much as possible the hoses length between distributors and actuators.

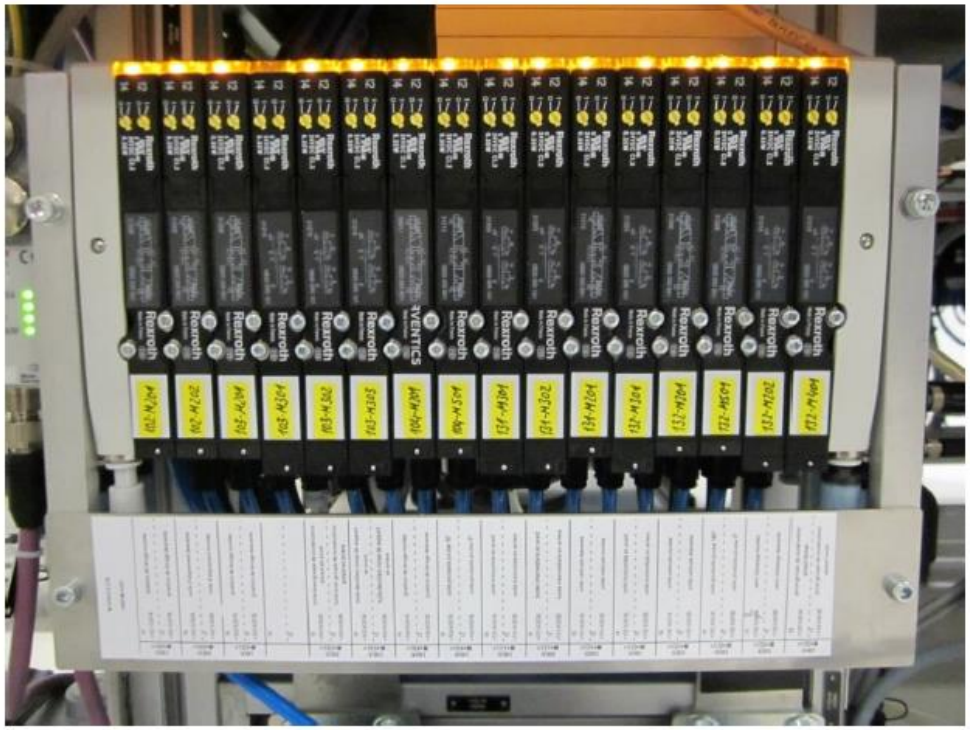
- Identification:

All distributors must be identified with labels in French placed on the distributor and on a stationary holder nearby.

The marking must match the electrical diagram.

Caution: Identify with permanent signs all the elements (actuators and hoses) remaining under pressure after a power switch-off (emergency stop).

Example of identification:



Cylinders and accessories:

- Cylinders

Use the product families PRA, MNI and CCI with magnetic sensors (even if the sensor is not used)

- Magnetic sensors

Use the sensors with threaded connector and PNP output circuit

- Fittings

Use the fittings series with quick connection in the metric variant.

- Flow throttling valves

Use the series with quick connection in the metric variant.
Caution: Use throttling valves with flow control on the air exhaust system (except when the technology requires regulation at the intake).

They must be installed on the actuators or as near as possible in order to guarantee the maximal efficiency. Their adjustment must be easy and be done safely.

Wiring:

The wiring must be executed with black polyurethane tube. The material must be adjusted to the used fittings and appropriate for the environment (oil, temperature, detergents, heavy mechanical stress...)
The tube Ø must be dimensioned so as to guarantee the right functioning of the installation.

The tubes remaining under pressure after the shutting-off of machine air must be red. Each tube must have an identical identification at both ends.

3. Electrical Standard

3.1 General information about electrical equipment:

- General characteristics

Compliance with French standard NF EN 60 204 – 1 “Electrical equipment of machines - Part 1: general requirements”
Compliance with European Machinery Directive 2006/42/EC
- Supply voltage: 400 V + 10% - Three-phase - 50 Hz TNs or TNc neutral system
- Control voltage: 24 VDC

- Display and interface

► Hours-run counter
► Cycle counter (OK parts)
► Total without reset to zero
► Shift with reset to zero
► Series with reset to zero

► Counter for non-conforming (NC) parts ⇒ display on the main control panel; it should be possible to program automatic machine shutdown in case of quality drifting: number of consecutive NC parts > x parts, % of NC parts > x%, etc…

► SPC counter (countdown) ⇒ display on the main control panel
► Cycle time

- Interior lighting: plan a LED interior lighting with switch on machine panel for operator or setter.

3.2 Electrical cabinet

3.2.1 Housing

- Double pickled steel sheet at least 2 mm thick:
  Maximal dimensions: Height: 2000 mm
  Depth: 600 mm
  Width: To discuss with the BOSCH Company.
- In a corrosive atmosphere, cabinets and boxes must be made of stainless steel.
- An elevation of 300 mm maxi is allowed for cable storage, which must be covered.
- Cabinet, elevation and duct in beige colour, RAL 7032.
- Functioning of all electrical devices must be guaranteed for an ambient temperature of 45 °C in the workshop.
- Upper part must be equipped with handling and hoisting features.
- All cabinet doors must have a double bit key lock.
- All doors giving access to a keyboard, a printer or something else (used by the production personnel) must be equipped with a single handle without lock.
- On the inside, the doors must be equipped with document holding compartments adjusted to document dimensions (binders format A4).
- Provide for a sliding shelf for browsing the documents in the cabinet.
- If electronics are used the cabinets must be equipped with air conditioning units (selection defined in the Requirements Specification), switching off when doors are open and with external condensate recovery (lateral mounting on door or wall, mounting on the roof is prohibited). In case of an electrical air conditioning unit, it must be connected to a socket protected by a 30 mA circuit breaker.

- The placing of the air conditioning unit and the direction of air flows must be optimised in order to cool the most sensitive devices.
- Plan a panel ventilation system with interchangeable filters.
- Cabinet temperature must be monitored and in case of a too high value, a warning must be displayed (Without machine shutdown).
- If the cabinet is not joined together with the machine, it must be removable by plugs.

- Special cables: Cables for numeric controls and measuring test points (linear encoders, coders, axes, buses, etc.) enter the cabinet through “cable duct connectors”.



Example of cable duct connectors

- Cabinets must have an interior lighting with LED automatically controlled by the doors opening; plan a 30 mA residual-current operated protective device at the starting point of this circuit, with its supply upstream of the disconnector.

3.2.2 Supply:

- 400 V three-phase 50 Hz – with single pole circuit breaker but earthing disconnector. In the French standard NFC15-100, neutral is active.
- If the cabinet is supplied with more than one voltage source (e.g. emergency or converter...) clearly identify in the cabinet the inlet points and the switch with an engraved riveted or crimped label.
- On the cabinet, put for each circuit an identified switch or circuit breaker, to be locked with 3 padlocks.
- If on the same machine, there are several cabinets equipped with a disconnector, the main cabinet must supply all other cabinets and be equipped with a red disconnector. The other cabinets must have a black disconnector.
- It should be possible to open the cabinet doors with the disconnector in ON position. The disconnector must remain in function and be controllable even with opened doors.
  Lateral or door mounting is to be favoured, **and devices with linkage are prohibited**.
- The holder (of disconnector, control light or other electrical equipment) must be directly connected to the protection conductor of the cabinet (cable cross-section must be at least equal to the cross-section of the cables supplying the electrical equipment)
- CAUTION! The switch-off efficiency of the protection devices must be 20 KA minimum.
- Control voltage 24 VDC for relays, inputs and outputs (sensors, electro-valves, etc…).
- Connect one of the transformer’s secondary polarities to the ground to avoid unfortunate malfunctions.
- Plan 1 socket (French standard) 2P+T (16A) in the cabinet and 1 socket (French standard) 2P+T (16A) on the panel or on the machine for programming devices (protected by 30 mA differential circuit breaker).
- After opening the machine’s switching-off device, only the cabinet lighting and the 2 sockets should remain powered.
- Plan external connectors for the connecting of programming and diagnosis consoles to the electrical devices (SUB-D connector, RJ 45, etc.)

- The wiring of elements before the disconnector must be put in yellow and orange cable sheaths.
- Identify and group (if possible) the devices with supplying before the disconnector.
- Motor starts and transformers primaries must be protected by magneto-thermic circuit breakers.
- The circuit breakers supply must be made from above: if not, signal it.
- Emergency stop complying with standards EN13850 and EN 60204-1
- On the machine terminal block (info by dry contact) provide for the possibility to perform end-of-cycle shutdowns and emergency stops (info by switching between 1 and 0) for implementing external installations as specified in the Requirements Specification (e.g. fire alarm, suction system, collecting tank, etc…).
- The cables transporting analogical signals must be shielded and earthed.
- The buses cables as well as those for measuring and computing devices must be flexible and their pathway must be separated from the power circuit.

3.2.3 Easy maintenance

- Occupancy or fill rate in the cabinet
  Plan a reserved space of approx. 30 % for potential extension (global space, cable duct, terminal blocks, PLC frame and NC, cable bushings ports, etc.).
  Connect all PLC inputs/outputs to the terminal blocks (even if not used)

- Accessibility
  All electrical equipment must be placed so as to enable easy checking, repairing and replacing.
  Position the equipment upright.
  Where possible, each device electrically positioned downstream of another one should be mounted physically underneath it.
  Group the devices by type: power circuit, control circuit, etc…
  Connecting of inlet conductors must be made on the device upper terminals (if not, provide for a marking on the device).
  Plan sufficient space (100 mm ventilation) between the ducts and the electronic devices. Cables should not be placed on the device.
  Provide for sufficient cable length on the terminal blocks as well as for sufficient distance to the ducts in order to allow for the maintenance works (use of current clamps, replacement of cable ends...).
  Clearly readable markings.
  Connection wiring between electrical cabinet and installation ⇒ industrial connector similar to HARTING connectors for intensive use.
  Guarantee the accessibility to the devices securing screws.

3.2.4 Identification

- Identification (numeric and alphanumeric) on the drawings must be consistent with the one put on the device.
- The devices remaining powered after opening the main switching-off device of the cabinet must be identified with a label in French (black on yellow background).
- Use the light blue colour for neutral conductor.
- Colours of cables, control buttons and warning lights must comply with the standard EN 60204-1.
- The cables coming from another cabinet which remain powered when the disconnector is opened must be wired with orange cables (e.g. Interconnection inputs/outputs between cabinets).

3.3 Machine

3.3.1 Inputs

- Inductive proximity sensors

Sensors with threaded connection or with cable with remote connector.
PNP inductive sensors with diode (PUR sheath).
3 wires 12-30 VDC
Metallic sensor housing if possible.
In a corrosive atmosphere connectors must be made of stainless steel.
Use conventional sensors even with special cylinders.
Don't use sensors with extended operating distance.
Don't use multifunction sensors (PNP/NPN or NO NC configurable by wiring).

- Photocells / pressure switch

IP 66
PNP
With threaded connection or with cable with remote connector.
3 wires 12-30 VDC (or 4 wires if additional functions are required)



Example of protection

- Control elements

Emergency stop must be equipped with mechanical latching function, Ø 22 mm, and be placed so as to be accessible without risk of unintended actuation (not in a passageway); if not possible, it must be protected.

Warning lights on the panels have a 22 mm Ø and must be equipped with LED lamps.
Buttons on the panels have a 22 mm Ø.
Numeric and alphanumeric keyboards are allowed but with built-in mouse of the trackball type.
Key operated switches will be equipped with a SSG10 key (at least in duplicate)

- Capacitive sensors

Avoid this type of sensor unless it is required for a specific detection.

- Magnetic cylinder sensors

With threaded connector, 24 VDC with control lights.

3.3.2 Outputs (actuators)

- Distributors

With status indicators and connectors.
The distributors (power > 30 W continuous) must be supplied by static relays.

- Signal tower

  The signal tower must be placed in order to be visible from top to bottom and **equipped with LED lamps**

  - Red — Machine shutdown
  - Orange — Waiting for operator (SPC validation, consumables replacement...)
  - Blue — Series changeover
  - Green — Automatic mode
  - Colourless — Specific use (home position...)

- Step motors with connectors
  IP xxx (level to be determined according to environment).
- Axes motors
  IP xxx (level to be determined according to operation).
  Synchronous motors with connectors (ringless and brushless)
- Asynchronous motor
  Minimal insulation class B
  Minimal protection IP 55
  Metallic terminal box
  Metallic fan cover
  Select low-energy motors according to the valid class.
  Soft start if the power exceeds 5 kW on the machines.

3.3.3 Miscellaneous

- Place an engraved riveted or crimped label near each inlet element or actuator, mentioning the reference number on the drawing, and attach another identical label on the cable with a clamp.
- **For trolleys numeric controls, place a label mentioning the axis name (X, Y...), the work direction in relation to NC (+, -, as seen from the control panel) and the reference mark on each axis.**
- It is recommended to use remote input/output modules instead of distribution boxes with terminal block (allow for 30 % free space).
- Robots must be delivered with the possibility of arm calibration based on a mechanical reference (specific tools included in the delivery).
- For numeric controls provide for an external device (like USB port) for programs backups (PROCESS).
- Plan measuring points for test instruments in order to facilitate the diagnoses and some secondary checks required for the process.

3.4 Programming

- Supplier quotation must detail the selected technical solution, including the technology description and the programming language choice.
  (Example: OpCon for Rexroth technologies VPB40…).
- The machine programming must have 3 different operating modes:

1 - Manual mode

Access to the dangerous area is only possible in manual mode with key switch and request for door unlocking.

This mode enables the machine operation under specific safety conditions complying with the European Machinery Directive.

In particular:

- In manual, each movement must be executable separately and safely.
- Operating of mobile dangerous elements (identified in the risk analysis) should be authorised only under reinforced safety conditions (for example: safety: controlled speed and stop, step-by-step movement or other appropriate measure)
- Prohibit any potential dangerous movement that may occur due to intentional or unintentional action on the machine internal sensors.

During works in the machine, to avoid hazards, the movements with opened doors should be possible only with an enabling device (dead man's control principle) or with reduced force/speed.

Bypassing the safety elements is forbidden during adjusting, repairing and maintenance works.

2 - Automatic mode
Starting in automatic mode should be possible only after going to home position and then actuating the cycle start button. During this operating mode all the dangerous mobile elements are not accessible.

3 - Semi-automatic mode
To run the semi-automatic mode you have to actuate a step-by-step button; this mode facilitates the machine diagnosis and adjustment. During this operating mode all the dangerous mobile elements are not accessible.

- Failure diagnosis

Diagnosis general characteristics:
The HMI (Human machine interface) must signal the malfunctions and enable the progress status visualisation of each automatic graph. In any case, an efficient diagnosis must be suggested. If a failure is detected the green lamp on the light tower should not remain lit; such machine configuration will not be tolerated.

Failure type: The failures to be examined are of two types:

1. Failure causing a machine shutdown with no possibility to restart the production. This type must be clearly identified in order to repair the faulty element or technical function.
2. Failure causing a drop in performance regarding productivity and/or cycle time. This performance decrease is identified based on the productivity results and especially thanks to the cycle time (CT) displays. Most of the time this type of failure is not the complete malfunction of a technical element or function but a fault causing a drop in performance. The decreasing can be caused, for example, by movements' slowdowns, jamming, increased time-outs, lack of tools, etc.

Lastly, the industrial equipment must be programmed so as to be able to monitor a cycle time drift.
Develop a programming code imposing redundancy.

- Access level to the functions
  Accessibility: 3 access levels to the parameters on all automatic machines with NC, PC or programmable controller (PLC).
  Accessibility to the functions through these defined levels must be prioritised and blocked with password or key (see hereafter), and be of limited duration (adjustable time) for levels 2 and 3.

  - Level 1: operator
    ⇒ Auto cycle (not blocked)
    ⇒ Return to home position
    ⇒ Adjusting of dimensions
    ⇒ Tools replacements
    ⇒ Manual mode
    ⇒ Access to alarms
    ⇒ Failure diagnosis "shutdown"
    ⇒ Failure diagnosis "performance decrease"
  - Level 2: setter
    ⇒ Program changing (blocked)
    ⇒ Machine calibration
    ⇒ Creation of the parts programs (level to be discussed with Bosch Company during the conception)
    ⇒ SPC parameters
  - Level 3: developer and maintenance personnel (blocked)

⇒ Creation of the parts programs
⇒ Access to all functions (machine configuration, programs zeroing for diagnosis, CTR parameters, machine operation, etc…)

For more details, please see the Machinery Directive 2006/42/EC part 1.2.5.

- Programming language
  - Computing:

Applications in C++ or C# and Visual Basic.
No programming with LabView.

- Automation:

Siemens equipment

Use of TIA Portal or Simatic manager. **No programming with S7 software and language SCL**.
Use WinCC or WinCC Flexible for HMI.

Rexroth equipment

Use OpCon 3.2 and HMI Modulo or newer version.
Latest version of Indraworks for the MTX numeric controls.

Schneider equipment

Use the software Unity-Pro and VIJEO Designer for the XBT, or newer versions.

3.5 Technical folder

It must comprise following documents:

- Electrical diagrams (**1 hard copy + 1 electronic version**)
  These diagrams must be developed with the software EPLAN 08 (modifiable sources); if other software programs should be used the previous approval of Bosch Company is mandatory. A pdf version must also be delivered.
  Diagrams of all supplementary equipment (e.g. robot cabinet, loading system, etc.) must be included in the main diagram with page references between diagrams.
- Program organisation (modifiable programs source codes on digital medium): Sequential function chart Grafcet or flow chart **with comments in French**.
- Programs edition (1 electronic version) with comments in French: symbolic charts of inputs, outputs, memento, time-out, etc…
- Technical folder (1 hard copy + 1 electronic version) for all electrical devices (converter, panel, sensor, door limit switch, safety box, etc…)
- Complete list of all electrical devices and equipment used on the machine and in the electrical cabinet (references, manufacturer...) **in the same document**.
- **All programs must be delivered with the source files and the comments**
  - **Source codes of the HMI program**
  - **Source codes of the programmable controller (PLC) program**
  - **Source codes of the NC program**
  - **Source codes of the PC program, for example: OpCon by ATMO**
- Backups for all programs (PLC, NC, etc.) must be delivered on data storage device, when the machine is placed on site and is in working order.
- **For all electrical devices** (cells, pressure switch, converter, etc.) you must deliver
  - the parameter setting procedure
  - the backup of parameters and, if necessary, of program, on data media or as hard copy
  - the name of the used software and its version
  - the loading/unloading procedures for the data in the device
  - the IP address (if using a network cable) and the address setting procedure.
- Deliver the quality survey for bus signals (example: Software Profiview for profibus) at the time of machine installation on site.
- A thermographic inspection of the electrical cabinet will be performed on site, on start-up, in order to check the compliance with the planned terms of use.
- For all electromechanical, electro-hydraulic and electro-pneumatic systems deliver the setting procedure as well as the adjustment procedure used in case of replacement.
- Deliver the setting procedure performed after the replacement of an axis motor, a linear encoder or a coder.

3.6 Electrical equipment

Description	Brand **for RzP**	Type
Contactors	Schneider, Siemens	
Secondary contactors	Schneider, Siemens	
Pluggable relays	Schneider, Schrack, Murrektronic	
Circuit breakers	Schneider, Siemens	Pluggable devices are not authorised.
Control element (push-button switch, control light...)	Schneider, Siemens	
Control box	Schneider, Siemens	
Cabinet	Rittal, Schneider	Double bit key type AE/PS
Control box and panel	Rose, Rittal, Schneider	Double bit key type CP
Position switch (limit switch)	Schneider, Siemens	
Inductive sensors	Schneider, IFM, Balluff, Schunk, Rexroth	
Photo-electrical sensors	IFM, Balluf , Sick, Keyence	
Magnetic sensors	Bosch Rexroth , Balluf, SMC, Schunk	
I/O dispatcher	Bosch Rexroth , Lumberg ,Siemens	
Pressure switch	Bosch, Schneider, IFM, Herion	
Flow meter	Bamo , Burkert, IFM	
Levels	Endress-Hauser, Jola, Bamo	
Linear encoder	Heidenhain	
Coder	Balluff, Heidenhain	
Signal tower	Schneider, Balluf	
Safety modules	Schneider, Pilz	
Terminals	Weidmüller, Legrand	
Switch disconnector	Schneider, Kraus	
Circuit breaker disconnector	Naimer, Socomec	
Air conditioning unit	Rittal	SK 3293-100 / SK 3290-100 (after approval)
Keyboard with display	Bosch, Lauer, Schneider, Sutron, Siemens	
Keyboard alone	Schneider, Rexroth	
Display alone	Lauer, Schneider, Siemens, Rexroth	
Brushless motor, servomotor	Rexroth	
Machine lighting	Waldmann, BOSCH	
Light barrier	Jay Electronique, Leuze Télémécanique	
T° indicator and controller	PMA , Eurotherm	
Cycle time display		
Numeric control	Rexroth, Siemens, Fanuc	
Converter for axis and spindle	Rexroth, Siemens	
Axis and spindle motor	Bosch, Siemens	
Asynchronous motor	Leroy-Sommer or similar	Standardised dimensions and fixations
		Classification: IE3 according to standard IEC 60034-30
Asynchronous motor converter	Schneider, Siemens, Rexroth	
Safety switch	Siemens, Euchner, Schneider	
Modular interface	Lutze, Murrelectronick, Entrelec,	
Time or pulse counter (48 x 48 mm)	Crouzet, Syrelec or Hengstler	
Robot	Stäublï, APAS, KUKA	
PLC	Rexroth, Siemens, Schneider	
Transformer	Legrand	
DC Supply	Siemens, Murrelectronic, Legrand	
Identification cables, measuring cable (on machine)	SES, Legrand, Movomatic	

4. Computer Security, Network Data:

For all used hardware, the following elements must be delivered to the Bosch Company on pre-acceptance:

- Manufacturer name
- Model
- Series number
- Inventory number
- Operating system (if possible on tendering)
- Hardware specifications
- Specific software

Operating system should not be outdated.

Servers and databases are exclusively provided by the Bosch Company.

The manufacturer must offer to make the machine data available based on an OPC type server (motor current, motor temperature, pressure, etc.).

Access to all data media (USB …) must be blocked. The use of such media to collect data on a regular basis is prohibited.

The Bosch Company must be able to equip all hardware with an antivirus. In exceptional cases, the Bosch Company can authorise the absence of antivirus but only if justified and approved by Bosch.

Systems updating must be possible. Otherwise, it has to be justified.

Connected user should not have access as an “administrator” and he should have the possibility of changing periodically his password.

If remote maintenance is planned for remote servicing or repairing of the WEq, this can only be performed according to the valid Bosch standard at the moment of the said maintenance, this solution having been tested and approved on WEq acceptance.

Computing equipment of the WEq will be obligatorily connected within partitioned networks with limited accesses.

Wi-Fi network will be accessible only in particular cases.

The needs for export and the associated formats must be submitted to the approval of the Bosch Company with the tendering.

5- Additional Requirements in relation to Security

Handling of the WEq and its components must be possible by means like handling trolleys, rollers, etc… without necessity to use suspension hoisting devices such as bridge cranes, cranes, etc… as the workshops at the Bosch plant in Rodez are not equipped with such devices. WEq hoisting and handling instructions must be sufficiently detailed in the user manual and be approved by the Bosch Company. Before delivery to the Bosch Company you have to provide a drawing of the WEq and its components mentioning the dimensions, masses, gravity centres and anchoring or gripping points.

Equipment must be designed so as to avoid the wearing of PPE by the users.

Use of products or substances that are not referenced in this Standard, for the WEq operation or maintenance, has to be approved by the Bosch Company beforehand, basing on the provided safety data sheets.

If planning the use of fluids, the manufacturer must design the WEq for a safe operation without hazards from filling, using, circulating, collecting and discharging.

Parts coming out of the WEq should not be hotter than 45°C.

The machine must be sufficiently isolated to guarantee a temperature lower than 50°C of all hot elements, in order to avoid contact burn risks.

Noise level

During a reference cycle, the noise level will not exceed 75 dB (A) in a 1 m distant zone around the WEq.

If this limit cannot be met, the supplier has to submit technical solutions to the prior approval of the Bosch Company.

The Bosch Company assesses the WEq fire risk and consequently, the fire protection system is defined and specified in the Requirements Specification.

The Bosch Company reserves the right to let a certification body perform a compliance test on pre-acceptance and/or on final acceptance of the equipment.

6- Additional Requirements in relation to Environment

WEq will be designed so as to avoid any environmental impact, thus minimising air and water emissions, generated waste, water and energy consumptions and use of chemicals.

Where impacts are unavoidable, the supplier must prove the use of the best available economical technology to restrict them.

In the case of air emissions the WEq will be equipped with a recycling or treatment system for these emissions coupled with measuring and control means.

Aside from being restricted in quantity, generated waste (for example wear parts...) must be compatible with optimal recovery (enhanced waste separation).

Use of environmental hazardous substances as defined in the applicable regulations is prohibited. If this type of substances is required the previous approval of Bosch Company is mandatory based on a documentary proof.

When using water:

The concerned WEq must be equipped with water counters (mains water as well as deionised water if relevant).

The counter must be connected at the WEq inlet and record all its water consumptions (filling, topping-up, tanks rinsing ...).

Note that the washing machines must be equipped with retention tanks according to applicable regulation; if a high level is detected the immediate cycle shutdown must be guaranteed as well as the closing of water inlet electro-valves.

7- **References**

MP-206.01	Procédure mise en production nouveaux Equipements de Travail (Procedure to introduce new Work Equipment into production)
CdC CP/MAE :	Liste cahier des charges CP/MAE (List Requirement Specification CP/MAE)

8. Approval Table

	Acronyms + Name
Editor (signatory)	RzP/TEF16 +RzP/TEF2
Approval (signatory)	RzP/TEF

2	Changement de nom du document : Passage de RzP-MD-22-07b à MP-206.01-912-RzP pour intégration dans BBM	Bauguil William MFE1	25/10/2023
1	Suite à un axe d'amélioration relevé au cours de l'audit interne de PS/QMM11 en 11/2019, ajout de ce suivi des modifications pour l'ensemble de la documentation RzP.	Roquigny Florent (PS/QMM1-RzP)	13/01/2020
Version	Modification, évolution du document	Rédacteur	Date